MASTER'S THESIS

[Working title]

[Optional descriptive subtitle]

[Your Name]

Supervisor: [Supervisor Name]

[Department], [University]

[Semester Year]

Abstract

Scaffold: replace this text with the abstract.

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Technical glossary **10**

List of Symbols and Notation

Vectors are columns. Symbols are defined at first use; the list collects the recurring notation.

Symbol	Meaning
$x \in \mathbb{R}^n$	Unknown vector.
$p \in \mathcal{P} \subseteq \mathbb{R}^m$	Parameter vector.
$F(x, p)$	Residual.
F_x	Jacobian of F with respect to x.

Acronyms

AD automatic differentiation

DAE differential-algebraic equation

ODE ordinary differential equation

Chapter 1

Introduction

1.1 Motivation

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Bibliography

[1] P. Kime, M. Wemheuer and P. Lehman. *The* ***biblatex*** *Package*. URL: https://ctan.org/pkg/biblatex.

[2] The LaTeX Project. *LaTeX – A document preparation system*. URL: https://www.latex-project.org/ (visited on 14/10/2026).

Appendix A

LaTeX examples

This appendix demonstrates the scaffold's conventions. It is illustrative material, not a claim about the thesis research. Remove it when writing.

A.1 Equations and definitions

Let $x \in \mathbb{R}^n$ be an unknown vector and $p \in \mathcal{P} \subseteq \mathbb{R}^m$ a parameter vector. Consider

$$F(x, p) = 0, \qquad F : \mathbb{R}^n \times \mathcal{P} \longrightarrow \mathbb{R}^n. \tag{A.1}$$

The residual F measures the mismatch in the equations. Its Jacobian with respect to x is written F_x. Equation (A.1) defines an implicit problem.

Definition A.1 (Euclidean norm)**.** For $x \in \mathbb{R}^n$, the Euclidean norm is

$$\|x\| = \sqrt{\langle x, x \rangle} = \left(\sum_{i=1}^{n} x_i^2 \right)^{1/2}.$$

Proposition A.2 (Zero norm)**.** *For $x \in \mathbb{R}^n$, $\|x\| = 0$ if and only if $x = 0$.*

Proof. If $\|x\| = 0$, then $\sum_i x_i^2 = 0$. Every summand is nonnegative, so each $x_i^2 = 0$ and hence $x = 0$. Conversely, substitution of $x = 0$ in definition A.1 gives $\|x\| = 0$. □

Remark A.3. A small residual alone does not guarantee a small error in the unknown. The relationship depends on conditioning and the scales of the equations.

The environments share a chapter counter; proposition A.2 and remark A.3 illustrate references to different statement types.

A.2 Acronyms and citations

A time-dependent model may take the form of an ordinary differential equation (ODE) or a differential-algebraic equation (DAE). Derivatives of computational expressions can be evaluated using automatic differentiation (AD). Later uses remain abbreviated: ODE, DAE, and AD. Definitions still belong in the prose, with the lists serving as reference aids.

For a citation example, The LaTeX Project [2] describes the LaTeX project. Bibliography formatting is provided by `biblatex`[1]. These documentation references are sample entries to replace with your literature.

A.3 An algorithm

For fixed p, algorithm A.1 illustrates a Newton iteration. It assumes each required Jacobian is nonsingular. The returned Boolean records whether the requested residual tolerance was achieved; it is not an error bound.

Algorithm A.1 Newton iteration for a fixed parameter vector

Require: Initial guess x_0, parameter p, tolerance $\varepsilon > 0$, iteration limit $K \geq 1$

$x \leftarrow x_0$
for $k = 0, \dots, K-1$ **do**
  if $\|F(x, p)\| \leq \varepsilon$ **then**
    return (x, true)
  end if
  Solve $F_x(x, p)s = -F(x, p)$ for s
  $x \leftarrow x + s$
end for
return $(x, \|F(x, p)\| \leq \varepsilon)$

A.4 Tables, figures and code

The numbers in table A.1 are illustrative formatting data. The table uses horizontal rules and decimal alignment.

Table A.1: Illustrative numerical values, not experimental results.

Case	Residual norm	Time (ms)
A	0.125	12.4
B	0.031	18.7

For a physical quantity, write $t = 2.5\,\text{ms}$. Figure files belong in `figures/`; the source below is a pattern to use once a real figure exists.

```
\begin{figure}[htbp]
  \centering
  \includegraphics[width=0.8\linewidth]{convergence.pdf}
  \caption{Explain what is plotted and why it matters.}
  \label{fig:convergence}
\end{figure}
```

A short source excerpt can be included directly:

```
def residual(x, p):
    return x*x - p
```

Listing A.1: A scalar residual used only as a syntax example.

Listing A.1 represents $F(x, p) = x^2 - p$.

Technical glossary

Jacobian The matrix representing the derivative of a vector-valued function with respect to its input coordinates

residual A function whose value measures the mismatch in a set of equations; a solution makes the residual zero